

Evolutionary Guided Decoding: Iterative Value Refinement for LLMs

进化引导解码：用于大语言模型的迭代价值优化

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Abstract

摘要

While guided decoding, especially value-guided methods, has emerged as a cost-effective alternative for controlling language model outputs without re-training models, its effectiveness is limited by the accuracy of the value function. We identify that this inaccuracy stems from a core distributional gap: existing methods train static value functions on trajectories sampled exclusively from the base policy, which inherently confines their training to a narrow and suboptimal view of the potential output space. We propose Iterative Value Refinement, a evolutionary framework designed to narrow this gap. It employs Value Exploration to provide a more comprehensive and robust training signal, complemented by Iterative Self-Refinement, which uses the improved value function from one iteration to guide the generation of higher-quality data for the next. Extensive experiments on text summarization, multi-turn dialogue, and instruction following demonstrate the effectiveness of our framework in aligning language models. Our approach not only achieves alignment but also significantly reduces computational costs by leveraging principled value function optimization for efficient and effective control.

在引导解码过程中，特别是基于价值的方法，已经成为一种在不重新训练模型的情况下控制语言模型输出的经济有效的替代方案。然而，其效果受到价值函数准确性的限制。我们发现这种不准确性源于一个核心的分布差距：现有方法在仅从基础策略中采样的轨迹上训练静态价值函数，这本质上将它们的训练限制在一个狭窄且次优的潜在输出空间视图中。我们提出了一个进化框架，称为迭代价值细化（Iterative Value Refinement），旨在缩小这一差距。该框架采用价值探索（Value Exploration）来提供更全面且稳健的训练信号，并通过迭代自我细化（Iterative Self-Refinement）进行补充，后者利用一次迭代中改进的价值函数来指导生成下一次迭代中更高质量的数据。我们在文本摘要、多轮对话和指令遵循任务上的广泛实验表明，我们的框架在对齐语言模型方面是有效的。我们的方法不仅实现了对齐，还通过利用原理性的价值函数优化显著降低了计算成本，从而实现高效且有效的控制。

1 Introduction

1 引言

Reinforcement Learning from Human Feedback (RLHF) (Christiano et al., 2017; Bai et al., 2022; Ouyang et al., 2022) has emerged as a widely adopted approach to align advanced language models with human values and task requirements (Wei et al., 2022; Achiam et al., 2023; Chao et al.,

2024; Su et al., 2024a). However, traditional RLHF methods like Proximal Policy Optimization (PPO) (Christiano et al., 2017; Ouyang et al., 2022) suffer from high computational costs and training instability (Zheng et al., 2023b; Rafailov et al., 2024),

从人类反馈中进行强化学习（Reinforcement Learning from Human Feedback, RLHF）（Christiano 等，2017；Bai 等，2022；Ouyang 等，2022）已成为一种广泛采用的方法，用于将先进的语言模型与人类价值观和任务需求对齐（Wei 等，2022；Achiam 等，2023；Chao 等，2024；Su 等，2024a）。然而，传统的 RLHF 方法，如近端策略优化（Proximal Policy Optimization, PPO）（Christiano 等，2017；Ouyang 等，2022）存在计算成本高和训练不稳定的问题（Zheng 等，2023b；Rafailov 等，2024）

(a) The Distributional Gap in Conventional Methods

(a) 传统方法中的分布差异

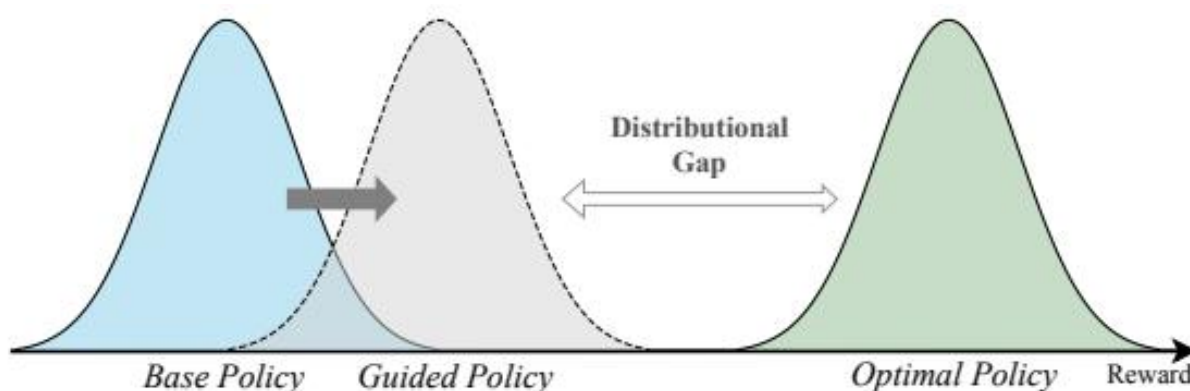


Figure 1: images/025573fb2a38d750d4cb0f0b53959aa539298323db19bdf431463c506b82d80c.jpg

(b) Our Iterative Value Refinement (IVR) Framework

(b) 我们的迭代价值优化（IVR）框架

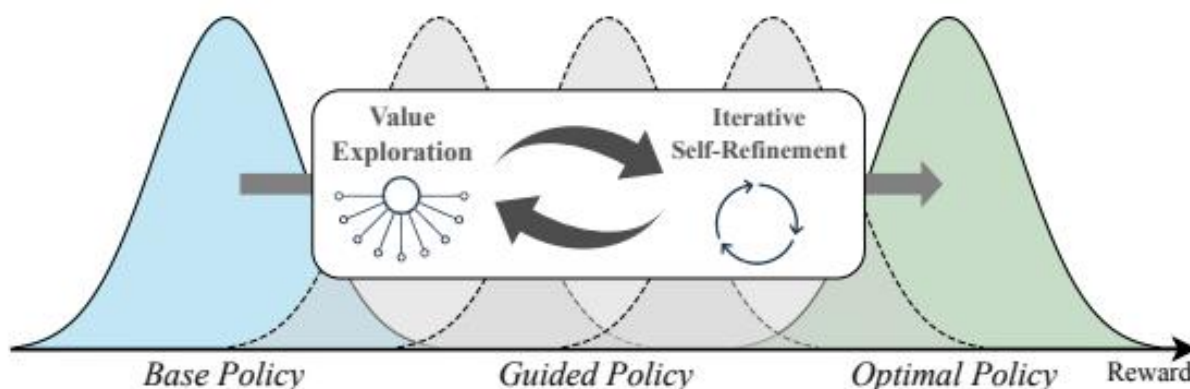


Figure 2: images/3ff31946e7fdd2015a8d2dfc3cab6f4c8b9a99fdfe0a9bdede4228bda95eaf7.jpg

Figure 1: (a) Conventional methods suffer from a distributional gap, limiting the guided policy from reaching optimal rewards. (b) Our Iterative Value Refinement framework narrows this gap through Value Exploration and Iterative Self-Refinement.

图 1: (a) 传统方法存在分布差距，限制了引导策略达到最优奖励的能力。(b) 我们的迭代价值细化框架通

过价值探索和迭代自我细化缩小了这一差距。

limiting their practicality for applications requiring flexible behavior control. Among various alternatives, guided decoding methods have gained increasing attention as they can steer model’s generation without expensive model re-training (Snell et al., 2022; Mudgal et al., 2023; Han et al., 2024; Chakraborty et al., 2024).

限制了它们在需要灵活行为控制的应用中的实用性。在各种替代方案中，引导解码方法逐渐受到越来越多的关注，因为它们可以在不进行昂贵的模型再训练的情况下引导模型的生成（Snell 等，2022；Mudgal 等，2023；Han 等，2024；Chakraborty 等，2024）。

Within this framework, value-guided approaches, which train a value function V_θ to evaluate partial outputs and steer the language model towards high-reward trajectories, have emerged as particularly promising (Yang and Klein, 2021; Qin et al., 2022; Mudgal et al., 2023; Han et al., 2024). Under the KL-regularized Reinforcement Learning framework, given an optimal value function V^* , they can derive a policy that maximizes expected rewards while maintaining a bounded KL-divergence from the base policy π_{base} .

在此框架下，以价值为导向的方法（value-guided approaches）逐渐显现为特别有前景的策略，这些方法通过训练一个价值函数 V_θ 来评估部分输出，并引导语言模型朝着高奖励轨迹发展（Yang 和 Klein, 2021; Qin 等，2022; Mudgal 等，2023; Han 等，2024）。在 KL-正则化强化学习框架下，给定一个最优价值函数 V^* ，它们可以推导出一种策略，在最大化预期奖励的同时，保持与基础策略 π_{base} 的 KL 散度在有界范围内。

As visualized in Figure 1(a), while guided decoding with an estimated value function can shift the base policy’s output distribution towards higher-reward regions, this improvement remains suboptimal compared to the theoretical maximum achiev-

如图 1(a) 所示，虽然使用估计的价值函数进行引导解码可以将基础策略的输出分布向高奖励区域转移，但这种改进与理论上的最大值相比仍显得不够理想。

able through the optimal value function V^* . We argue this suboptimality is rooted in the distributional gap between the learned value function and the optimal value function. Specifically, existing methods (Khanov et al., 2024; Mudgal et al., 2023) train a static value function on trajectories sampled exclusively from the base policy π_{base} , which inherently confines its training to a narrow and suboptimal view of the potential output space. The limitation leads to substantial suboptimality in value function estimation, ultimately hindering decoding effectiveness.

通过最优值函数 V^* 来实现。我们认为这种次优性源于学习到的值函数与最优值函数之间的分布差异。具体来说，现有方法（Khanov 等，2024；Mudgal 等，2023）仅在基于基础策略 π_{base} 采样的轨迹上训练静态值函数，这本质上限制了其训练范围，只能局限于潜在输出空间的一个狭窄且次优的视角。这一局限性导致值函数估计出现显著的次优性，最终阻碍了解码效果。

To mitigate this distributional gap, we introduce Iterative Value Refinement (IVR), as shown in Figure 1(b), a unified, self-bootstrapping framework that systematically improves the value function’s accuracy and robustness. It employs Value Exploration, which generates multiple trajectories per prompt to provide a more comprehensive and robust training signal for each update. This is complemented by Iterative Self-Refinement, where the learned value function from one iteration guides the generation of higher-quality data, progressively narrowing the distributional gap. Through this evolutionary refinement, IVR allows the value function to iteratively improve, enabling it to more accurately identify high-reward trajectories far beyond the scope of the initial base policy. Unlike traditional online RLHF methods that require repeatedly collecting preference data and retraining the policy model, IVR achieves policy improvement by optimizing only the value function, substantially reducing computational costs while maintaining the benefits of iterative refinement.

为了解决这一分布差距，我们引入了迭代价值细化（IVR），如图 1(b) 所示，这是一个统一的、自我引导的框架，系统地提高价值函数的准确性和鲁棒性。它采用价值探索，通过每个提示生成多条轨迹，以提供更全面且稳健的训练信号。这由迭代自我细化加以补充，其中一次迭代中学到的价值函数指导生成高质量的

数据，逐步缩小分布差距。通过这种进化式细化，IVR 使价值函数能够迭代改进，从而使其能够更准确地识别远超初始基础策略范围的高奖励轨迹。与传统的在线 RLHF 方法不同，后者需要反复收集偏好数据并重新训练策略模型，IVR 仅通过优化价值函数来实现策略改进，大大降低了计算成本，同时保持了迭代细化的优势。

Our main contributions are summarized as following:

我们的主要贡献总结如下：

- We introduce IVR, a novel framework that narrows the distributional gap in value function training by unifying Value Exploration and Iterative Self-Refinement, enabling the value function to be iteratively improved for greater accuracy and robustness.
- We demonstrate the generalizability and effectiveness of IVR by conducting extensive experiments across a variety of challenging tasks, including text summarization, multi-turn dialogue, and instruction following, showing consistent improvement in performance over existing approaches. Our method outperforms baseline methods in terms of reward scores across all evaluated tasks, which is further corroborated by a
- 我们引入了 IVR，一个新颖的框架，通过统一价值探索和迭代自优化，缩小价值函数训练中的分布差距，使价值函数能够被迭代改进，从而获得更高的准确性和鲁棒性。
- 我们通过在多种具有挑战性的任务上进行广泛的实验，展示了 IVR 的通用性和有效性，包括文本摘要、多轮对话和指令跟随等任务，结果显示在现有方法之上性能有持续提升。我们的方法在所有评估任务中都优于基线方法，在奖励得分方面表现更优，这一结果也得到了进一步的验证。

77.52% GPT-4 win rate on the Multi-turn Dialogue task against the base policy.

77.52% 的 GPT-4 在多轮对话任务中战胜基础策略。

- We conduct extensive empirical analysis on the impact of sampling trajectories and training iterations, providing practical insights for implementing value-guided decoding methods.

- 我们对采样轨迹和训练迭代的影响进行了广泛的实证分析，为实施基于价值引导的解码方法提供了实用的见解。

2 Preliminaries

2 预备知识

2.1 The Token-level Markov Decision Process for RLHF

2.1 基于标记的马尔可夫决策过程用于 RLHF

We define the text generation mechanism of large language models (LLMs) as a token-level Markov Decision Process (MDP). Define the tuple $\mathcal{M} = (\mathbf{S}, \mathbf{A}, f, \mathbf{R}, \rho)$, where $\mathbf{S}$ denotes the state space encompassing all previously generated tokens (i.e., $s_t = \{x_0, \dots, x_m, y_0, \dots, y_t\}$). Here, $x_0, \dots, x_m$ are tokens from the initial prompt $\mathbf{x}$, and $y_0, \dots, y_t$ are tokens generated by the model up to time t . The action space $\mathbf{A}$ represents the vocabulary of tokens. The function f , representing deterministic transitions between states, is defined as $f(s, a) = s \oplus a$, where $\oplus$ indicates concatenation and $a \in \mathbf{A}$. The initial state distribution ρ is defined over the prompts $\mathbf{x}$, with each initial state s_1 comprising the tokens from $\mathbf{x}$. $R : \mathbf{S} \times \mathbf{A} \rightarrow \mathbb{R}$ represents the token-level reward.

我们将大型语言模型 (LLMs) 的文本生成机制定义为一个基于标记的马尔可夫决策过程 (MDP)。定义元组 $\mathcal{M} = (\mathbf{S}, \mathbf{A}, f, \mathbf{R}, \rho)$ ，其中 $\mathbf{S}$ 表示状态空间，包含所有先前生成的标记（即 $s_t = \{x_0, \dots, x_m, y_0, \dots, y_t\}$ ）。在此， $x_0, \dots, x_m$ 是初始提示 $\mathbf{x}$ 中的标记， $y_0, \dots, y_t$ 是模型在时间 t 之前生成的标记。动作空间 $\mathbf{A}$ 表示标记的词汇表。函数 f ，表示状态之间的确定性转移，定义为 $f(s, a) = s \oplus a$ ，其中 $\oplus$ 表示连接操作， $a \in \mathbf{A}$ 。初始状态分布 ρ 定义在提示 $\mathbf{x}$ 上，每个初始状态 s_1 包含来自 $\mathbf{x}$ 的标记。 $R: \mathbf{S} \times \mathbf{A} \rightarrow \mathbb{R}$ 表示标记级别的奖励。

KL-regularized RL and Optimal Policy. The objective of KL-regularized RLHF can be formulated as the following optimization problem:

KL-正则化强化学习与最优策略。KL-正则化 RLHF 的目标可以表述为以下优化问题：

$$\max_{\pi} \mathbb{E}_{y \sim \pi} [\mathbf{R}(\mathbf{x}, \mathbf{y})] \quad s.t. \quad D_{KL}(\pi || \pi_{base}) < \epsilon, \quad (1)$$

where $D_{KL}(\pi || \pi_{base})$ denotes the KL divergence between the policy π and the base policy π_{base} .

其中 $D_{KL}(\pi || \pi_{base})$ 表示策略 π 与基准策略 π_{base} 之间的 KL 散度。

Following the prior line of works (Peters and Schaal, 2007; Peng et al., 2019), the closed form solution to the KL-regularized RL problem can be represented as:

遵循之前的系列工作 (Peters 和 Schaal, 2007; Peng 等, 2019)，KL-正则化强化学习问题的闭式解可以表示为：

$$\pi^*(y_{t+1} | \mathbf{x}, y_{\leq t}) \propto \pi_{base}(y_{t+1} | \mathbf{x}, y_{\leq t}) e^{\beta \mathbf{Q}^*(y_{t+1} | \mathbf{x}, y_{\leq t})}, \quad (2)$$

where β is a control parameter characterizing the trade-off between the reward and the KL divergence. Given the deterministic transition model f , the expected future reward after action y_{t+1} , or Q-value $\mathbf{Q}^*(y_{t+1} | \mathbf{x}, y_{\leq t})$, equates to the value of the subsequent state. The Q-value of taking a specific action y_{t+1} at the current state can be then transformed into the state value of their concatenation:

其中 β 是一个控制参数，用于表征奖励与 KL 散度之间的权衡。给定确定性的转移模型 f ，执行动作 y_{t+1} 后的预期未来奖励，或 Q 值 $\mathbf{Q}^*(y_{t+1} | \mathbf{x}, y_{\leq t})$ ，等于后续状态的值。当前状态下采取特定动作 y_{t+1} 的 Q 值可以转换为它们的连接状态值：

$$\mathbf{Q}^*(y_{t+1} | \mathbf{x}, y_{\leq t}) = \mathbf{V}^*(\mathbf{x}, y_{\leq t} \oplus y_{t+1}) = \mathbf{V}^*(\mathbf{x}, y_{\leq t+1}). \quad (3)$$

Thus, the optimal policy in Equation 2 is rewritten as

因此，方程 2 中的最优策略可以重写为

$$\pi^*(y_{t+1} | \mathbf{x}, y_{\leq t}) \propto \pi_{base}(y_{t+1} | \mathbf{x}, y_{\leq t}) e^{\beta \mathbf{V}^*(\mathbf{x}, y_{\leq t+1})}. \quad (4)$$

2.2 LLM Alignment via Value-Guided Search

2.2 基于价值引导的搜索进行大语言模型对齐

The problem of LLM alignment can be formally defined as solving for the optimal decoding policy π^* under the token level MDP $\mathcal{M}$ in Equation 4. Here, we present two search strategies for

decoding.

LLM 对齐问题可以形式化地定义为在公式 4 中的 token 级 MDP M 下求解最优解码策略 π^* 。在这里，我们提出了两种解码搜索策略。

Value Guided Top-k Sampling. A prevalent strategy to address the computational expense of calculating values for all possible subsequent tokens is to compute values only for the top-k tokens as determined by the base policy at each step:

值引导的 Top-k 采样。一种普遍采用的策略是，为了解决计算所有可能后续 token 的值所带来的计算开销，可以在每一步仅计算由基础策略确定的前 k 个 token 的值：

$$\pi(y_{t+1}|\mathbf{x}, y_{\leq t}) \propto \begin{cases} \pi_{\text{base}}(y_{t+1}|\mathbf{x}, y_{\leq t})e^{\beta \mathbf{V}^*(\mathbf{x}, y_{\leq t+1})} & y_{t+1} \in \text{top-k} \\ \pi_{\text{base}}(y_{t+1}|\mathbf{x}, y_{\leq t})e^{\beta \mathbf{V}^*(\mathbf{x}, y_{\leq t})} & y_{t+1} \notin \text{top-k} \end{cases} \quad (5)$$

In our pilot experiments, we found that the value-guided sampling can be further simplified by perturbing the distribution at the block-level instead of the token-level. This modification strikes a better balance between performance and efficiency, preserving effectiveness while enhancing inference speed. For more details, please refer to Appendix A.

在我们的试点实验中，我们发现通过在块级而非 token 级扰动分布，可以进一步简化值引导的采样。这种修改在性能和效率之间取得了更好的平衡，在保持效果的同时提高了推理速度。如需更多细节，请参阅附录 A。

Value Guided Blockwise Beam Search. Without considering the KL constraint, we propose value guided blockwise beam search to leverage the value function for decoding-time alignment. At each step, for each of the B candidate sequences, we sample B continuation blocks $y_{t:t+b}$ and rank all B^2 sequences according to $V_{\theta}(\mathbf{x}, y_{\leq t} \oplus y_{t:t+b})$. The top B sequences are retained as candidates for the next iteration until generation is complete, after which the sequence with the highest value score is selected as the output.

值引导的块状束搜索。在不考虑 KL 约束的情况下，我们提出值引导的块状束搜索方法，以利用价值函数进行解码时的对齐。在每一步中，对于每个 B 个候选序列，我们采样 B 个延续块 $y_{t:t+b}$ ，并根据 $V_{\theta}(\mathbf{x}, y_{\leq t} \oplus y_{t:t+b})$ 对所有 B^2 序列进行排序。保留排名前 B 的序列作为下一轮的候选，直到生成完成，之后选择价值得分最高的序列为输出。

3 Methodology

3 方法论

Challenges of Training Optimal Value Function. A significant challenge in implementing value-guided sampling is the necessity of accessing the optimal value function $\mathbf{V}^*(s_t)$ for each state s_t . This function denotes the maximum expected reward that can be achieved from state $s_t = (\mathbf{x}, y_{\leq t})$ when following the optimal policy π^* , until a complete answer y is generated. The function is defined as:

训练最优价值函数的挑战。在实现价值引导采样的过程中，一个显著的挑战是需要访问每个状态 s_t 的最优价值函数 $\mathbf{V}^*(s_t)$ 。该函数表示在遵循最优策略 π^* 的情况下，从状态 $s_t = (\mathbf{x}, y_{\leq t})$ 开始直到生成完整的答案 y 所能获得的最大预期奖励。该函数定义为：

$$\mathbf{V}^*(\mathbf{x}, y_{\leq t}) = \mathbb{E}_{\mathbf{y} \sim \pi^*(\cdot | \mathbf{x}, y_{\leq t})} \mathbf{R}(\mathbf{x}, \mathbf{y}). \quad (6)$$

In practice, $\mathbf{V}^*(\mathbf{x}, y_{\leq t})$ remains inaccessible, as it relies on the trajectory produced by the unattainable optimal policy π^* . Existing methods (Khanov et al., 2024; Mudgal et al., 2023) employ the base policy π_{base} as an approximation for π^* to estimate the optimal value function. However, these approaches often yield significant suboptimality due to the distributional gap between π_{base} and π^* . Specifically, training the value function exclusively on trajectories sampled from the base policy π_{base} confines its learning to a narrow and suboptimal view of the potential output space. This fundamental limitation leads to substantial suboptimality in value function estimation, as the learned function fails to accurately evaluate states that lie outside the base policy’s typical trajectory distribution, ultimately hindering the effectiveness of guided decoding.

实际上, $\mathbf{V}^*(\mathbf{x}, y_{\leq t})$ 仍然无法访问, 因为它依赖于不可实现的最优策略 π^* 产生的轨迹。现有方法 (Khanov 等, 2024; Mudgal 等, 2023) 使用基础策略 π_{base} 作为 π^* 的近似, 以估计最优值函数。然而, 这些方法由于 π_{base} 和 π^* 之间的分布差异, 常常导致显著的次优性。具体而言, 仅在从基础策略 π_{base} 采样的轨迹上训练值函数, 会限制其学习范围, 使其只能接触到潜在输出空间中狭窄且次优的视角。这一根本性限制导致值函数估计出现显著的次优性, 因为所学习的函数无法准确评估超出基础策略典型轨迹分布的状态, 最终阻碍了引导解码的有效性。

Overview. In this paper, we introduce Iterative Value Refinement to mitigate the distributional gap between the estimated value function and the optimal value function for guided decoding (refer to Algorithm 1 for the complete process). Our approach comprises two key components that work synergistically within a unified framework. First, we introduce Value Exploration, which expands the search space through multi-trajectory sampling to provide a more comprehensive and robust training signal for value function estimation. Second, we propose Iterative Self-Refinement, which leverages the improved value function from last iteration to guide the generation of higher-quality trajectories for the next iteration. This iterative process progressively narrows the distributional gap by enabling the value function to explore and learn from increasingly high-reward regions of the output space, thereby enhancing response quality during decoding.

概述. 在本文中, 我们引入了迭代价值细化 (Iterative Value Refinement), 以缓解估计价值函数与最优价值函数之间的分布差异, 用于引导解码 (完整过程请参见算法 1)。我们的方法包含两个关键组成部分, 它们在统一的框架内协同工作。首先, 我们引入了价值探索 (Value Exploration), 通过多轨迹采样扩展搜索空间, 从而为价值函数估计提供更全面且稳健的训练信号。其次, 我们提出了迭代自我细化 (Iterative Self-Refinement), 利用上一轮次改进后价值函数来指导生成下一轮次的高质量轨迹。这一迭代过程通过使价值函数能够探索并从输出空间中越来越高的奖励区域进行学习, 逐步缩小分布差异, 从而在解码过程中提升响应质量。

3.1 Iterative Value Refinement

3.1 迭代价值细化

Our iterative value refinement framework unifies Value Exploration and Iterative Self-Refinement into a cohesive approach that systematically improves value function accuracy through progressive refinement. The two components work in tandem: Value Exploration provides comprehensive training signals at each iteration, while Iterative Self-Refinement ensures that each subsequent iteration operates on higher-quality data than the previous one.

我们的迭代价值优化框架将价值探索和迭代自我优化统一为一种连贯的方法, 通过逐步优化系统性地提高价值函数的准确性。这两个组件协同工作: 价值探索在每次迭代中提供全面的训练信号, 而迭代自我优化则确保每次后续迭代所使用的数据质量高于前一次。

Value Exploration. We introduce Value Exploration, which utilizes stochastic sampling to im-

价值探索。我们引入价值探索，它利用随机采样来进

prove the accuracy of value function estimation by exploring a wider range of possible trajectories. Specifically, for a given prompt $\mathbf{x}$, we generate multiple outputs by performing stochastic sampling with the current policy. These outputs are then evaluated using the reward model R , which reflects alignment with human preferences.

通过探索更广泛的可能轨迹来验证价值函数估计的准确性。具体来说，对于给定的提示 $\mathbf{x}$ ，我们使用当前策略进行随机采样生成多个输出。这些输出随后通过奖励模型 R 进行评估，该模型反映了与人类偏好的一致性。

By sampling several trajectories and collecting their corresponding rewards, we can effectively train our value function with a more comprehensive and robust training signal. For each state $s_t = (\mathbf{x}, y_{\leq t})$ in the trajectory y , the estimated value for the current state is defined as:

通过采样多个轨迹并收集其对应的奖励，我们可以有效地使用更加全面且稳健的训练信号来训练我们的价值函数。对于轨迹 y 中的每个状态 $s_t = (\mathbf{x}, y_{\leq t})$ ，当前状态的估计值定义为：

$$\tilde{\mathbf{V}}^*(\mathbf{x}, y_{\leq t}) = \mathbf{R}(\mathbf{x}, y). \quad (7)$$

We then optimize $V_\theta(\mathbf{x}, y_{\leq t})$, parameterized by θ , to match $\hat{\mathbf{V}}^*(\mathbf{x}, y_{\leq t})$ using the following L_2 objective function:

我们随后优化 $V_\theta(\mathbf{x}, y_{\leq t})$ ，该参数由 θ 参数化，使用以下 L_2 目标函数来匹配 $\hat{\mathbf{V}}^*(\mathbf{x}, y_{\leq t})$ ：

$$\ell^*(\mathbf{x}, y; \theta) = \mathbb{E}_{\mathbf{x} \sim \mu} \left[\frac{1}{2} \sum_{t \in [|y|]} (V_\theta(\mathbf{x}, y_{\leq t}) - \mathbf{v}^*(\mathbf{x}, y_{\leq t}))^2 \right], \quad (8)$$

where μ is a distribution over training prompts.

其中 μ 是对训练提示的分布。

Iterative Self-Refinement. While Value Exploration enhances the exploration of potential trajectories through multi-trajectory sampling, the coverage of these trajectories is still inherently constrained by the current policy. To more effectively address the distributional gap from the optimal policy π^* , we propose Iterative Self-Refinement for progressively improving the value function. This strategy is founded on the principle that the optimized policy for the RL objective can be formulated as a value guided policy:

迭代自我优化。虽然价值探索通过多轨迹采样增强了潜在轨迹的探索，但这些轨迹的覆盖范围仍然受到当前策略的固有限制。为了更有效地解决最优策略 π^* 的分布差距，我们提出迭代自我优化策略，以逐步改进价值函数。该策略基于这样一个原则：用于强化学习目标的优化策略可以表述为一个价值引导的策略：

$$\pi_{V_\theta}(y_{\leq t+1} | \mathbf{x}, y_{\leq t}) \propto \pi_{\text{ref}}(y_{\leq t+1} | \mathbf{x}, y_{\leq t}) e^{\beta V_\theta(\mathbf{x}, y_{\leq t+1})}. \quad (9)$$

We then collect higher-quality trajectories by sampling from the policy π_{V_θ} , which is guided by the current value function V_θ . This sampling process can be represented as follows:

我们随后通过从当前价值函数 V_θ 引导的策略 π_{V_θ} 中进行采样，收集更高质量的轨迹。这一采样过程可以表示如下：

$$\hat{\mathbf{y}} \sim \pi_{V_\theta}(\cdot | \mathbf{x}), \quad (10)$$

where $\hat{y}$ denotes the complete trajectory sampled from the optimized policy π_{V_θ} , as defined in Equation 9.

其中 $\hat{y}$ 表示从优化策略 π_{V_θ} 中采样的完整轨迹，如公式 9 所定义。

With these higher-quality trajectories, we apply Value Exploration as outlined above. For each state $s_t = (\mathbf{x}, \hat{y}_{\leq t})$ in the sampled trajectory, we estimate its value as:

通过这些更高质量的轨迹，我们应用如上所述的价值探索。对于采样轨迹中的每个状态 $s_t = (\mathbf{x}, \hat{y}_{\leq t})$ ，我们估计其价值为：

$$\tilde{\mathbf{V}}^*(\mathbf{x}, \hat{y}_{\leq t}) = \mathbf{R}(\mathbf{x}, \hat{\mathbf{y}}). \quad (11)$$

Algorithm 1 Iterative Value Refinement

算法 1 迭代值优化

Input: reward model R , base model π_{ref} , training dataset μ Output: Value-guided policy π_{V_θ} , value function V_θ Initialize value function V_θ with pre-trained language model repeat - Step 1: Collect Multiple Trajectories - for $\mathbf{x}$ in μ do Sample K trajectories $\{\mathbf{y}_k\}_{k=1}^K \sim \pi_{base}(\cdot|\mathbf{x})$ Compute rewards $\{r_k = \mathbf{R}(\mathbf{x}, \mathbf{y}_k)\}_{k=1}^K$ end for - Step 2: Train the Value Function - For each trajectory state $s_t = (\mathbf{x}, y_t)$, set $\tilde{\mathbf{V}}^*(s_t) = \mathbf{R}(\mathbf{x}, \mathbf{y})$ Optimize V_θ using $L^*(\theta)$, where: $L^*(\theta) = \mathbb{E}_{\mathbf{x} \sim \mu} [\frac{1}{2} \sum_{t \in [y]} (V_\theta(\mathbf{x}, y_{\leq t}) - \tilde{\mathbf{V}}^*(\mathbf{x}, y_{\leq t}))^2]$ - Step 3: Policy Optimization - Define value-guided policy: $\pi_{V_\theta}(y_{t+1}|\mathbf{x}, y_{\leq t}) \propto \pi_{base}(y_{t+1}|\mathbf{x}, y_{\leq t})e^{\beta V_\theta(\mathbf{x}, y_{t+1})}$ Update policy: $\pi_{base} \leftarrow \pi_{V_\theta}$ until convergence

输入: 奖励模型 R , 基础模型 π_{ref} , 训练数据集 μ 输出: 值引导策略 π_{V_θ} , 值函数 V_θ 使用预训练语言模型初始化值函数 V_θ 重复 - 步骤 1: 收集多个轨迹 - 对于 $\mathbf{x}$ 在 μ 中: 采样 K 条轨迹 $\{\mathbf{y}_k\}_{k=1}^K \sim \pi_{base}(\cdot|\mathbf{x})$ 计算奖励 $\{r_k = \mathbf{R}(\mathbf{x}, \mathbf{y}_k)\}_{k=1}^K$ 结束循环 - 步骤 2: 训练值函数 - 对于每个轨迹状态 $s_t = (\mathbf{x}, y_t)$, 设置 $\tilde{\mathbf{V}}^*(s_t) = \mathbf{R}(\mathbf{x}, \mathbf{y})$ 使用 $L^*(\theta)$ 优化 V_θ , 其中: $L^*(\theta) = \mathbb{E}_{\mathbf{x} \sim \mu} [\frac{1}{2} \sum_{t \in [y]} (V_\theta(\mathbf{x}, y_{\leq t}) - \tilde{\mathbf{V}}^*(\mathbf{x}, y_{\leq t}))^2]$ - 步骤 3: 策略优化 - 定义值引导策略: $\pi_{V_\theta}(y_{t+1}|\mathbf{x}, y_{\leq t}) \propto \pi_{base}(y_{t+1}|\mathbf{x}, y_{\leq t})e^{\beta V_\theta(\mathbf{x}, y_{t+1})}$ 更新策略: $\pi_{base} \leftarrow \pi_{V_\theta}$ 直到收敛

The value function V_θ is then optimized by minimizing the following loss function:

价值函数 V_θ 然后通过最小化以下损失函数进行优化：

$$L^*(\theta) = \mathbb{E}_{\mathbf{x} \sim \mu} \left[\frac{1}{2} \sum_{t \in [\hat{\mathbf{y}}]} (V_\theta(\mathbf{x}, \hat{y}_{\leq t}) - \tilde{\mathbf{V}}^*(\mathbf{x}, \hat{y}_{\leq t}))^2 \right]. \quad (12)$$

This process can be repeated iteratively, with each iteration using the improved value function to guide the generation of higher-quality data for further training. The synergy between Value Exploration and Iterative Self-Refinement enables our framework to progressively narrow the distributional gap and achieve more accurate value function estimation.

这个过程可以迭代重复进行，每次迭代都使用改进的价值函数来指导生成更高质量的数据以进行进一步的训练。价值探索与迭代自优化之间的协同作用使我们的框架能够逐步缩小分布差异，并实现更准确的价值函数估计。

3.2 Why IVR Narrows the Distributional Gap

3.2 为什么 IVR 缩小了分布差距

We provide an informal discussion of why IVR narrows the distributional gap, framed through the notion of coverage between the data-collecting distribution and the visitation measure of the optimal policy.

我们以数据收集分布与最优策略的访问度量之间的覆盖概念为框架，非正式地讨论了 IVR 为何能够缩小分布差距。

We use the standard visitation measure of policy π :

我们使用标准的访问度量政策 π :

$$d^\pi(s, a) = \mathbb{E}_{s_1 \sim \rho} \left[\sum_{h=1}^H \mathbb{P}(s_h = s, a_h = a \mid s_1) \right], \quad (13)$$

which captures how often the state-action pair (s, a) is visited when following π from initial states $s_1 \sim \rho$.

这表示在从初始状态 $s_1 \sim \rho$ 跟随 π 时，状态-动作对 (s, a) 被访问的频率。

A recurring insight from offline and iterative RLHF is that the optimality gap between any approximate policy $\hat{\pi}$ and the optimal policy π^* is governed by how well the data-collecting distribution covers the visitation measure d^* of π^* (Xiong

从离线和迭代的 RLHF 中反复得出的见解是，任何近似策略 $\hat{\pi}$ 与最优策略 π^* 之间的最优性差距，由数据收集分布覆盖 π^* 的访问度量 d^* 的程度所决定。

et al., 2024; Ye et al., 2024). When training data is sampled from a distribution far from d^* , the value function cannot accurately evaluate states encountered under π^* , leading to a large optimality gap.

et al., 2024; Ye et al., 2024). 当训练数据是从与 d^* 相差很远的分布中采样时，价值函数无法准确评估在 π^* 下遇到的状态，从而导致较大的最优性差距。

IVR narrows this gap through two complementary mechanisms. Value Exploration broadens the support of the data-collecting distribution by sampling multiple trajectories per prompt rather than relying on a single rollout, providing wider coverage at each iteration. Iterative Self-Refinement progressively replaces the data-collecting policy π_{base} with the value-guided policy π_{V_θ} , whose visitation measure is biased toward higher-reward regions and therefore closer to d^* . Together, these two mechanisms iteratively shrink the coverage gap between $d^{\hat{\pi}}$ and d^* , which in turn reduces the optimality gap of the learned value function.

IVR 通过两种互补的机制来缩小这一差距。价值探索 (Value Exploration) 通过为每个提示采样多条轨迹，而不是依赖于单一的 rollout，从而扩大了数据收集分布的支持范围，在每次迭代中提供更广泛的覆盖。迭代自我优化 (Iterative Self-Refinement) 逐步用受价值引导的策略 π_{V_θ} 替代数据收集策略 π_{base} ，该策略的访问度量偏向于高奖励区域，因此更接近 d^* 。这两种机制共同作用，迭代地缩小 $d^{\hat{\pi}}$ 和 d^* 之间的覆盖差距，从而减少所学价值函数的最优性差距。

3.3 Connection to Online RLHF

3.3 与在线强化学习对齐（Online RLHF）的连接

Existing research indicates that online iterative RLHF can significantly enhance model performance (Xiong et al., 2024; Dong et al., 2024; Ye et al., 2024). In contrast to traditional online RLHF, which involves continuously collecting new preference data from the latest policy and retraining the policy model, our method eliminates the need for retraining. Instead, IVR focuses solely on optimizing the value function and employing guided decoding to iteratively improve the policy, thereby conserving computational resources.

现有研究表明，在线迭代强化学习反馈（RLHF）可以显著提升模型性能（Xiong 等，2024；Dong 等，2024；Ye 等，2024）。与传统的在线 RLHF 不同，后者需要持续从最新策略中收集新的偏好数据并重新训练策略模型，而我们的方法则无需重新训练。相反，IVR 仅专注于优化价值函数，并通过引导解码来迭代改进策略，从而节省计算资源。

4 Experiments

4 实验

Experimental Setup. For the summarization and multi-turn dialogue tasks, we first establish a base policy by supervised fine-tuning a pre-trained language model on the respective datasets to acquire basic task-specific capabilities and desired behaviors. For the instruction following task, we directly utilize publicly available instruction-tuned models. In all experiments, we parameterize the value function as a pre-trained language model backbone with a linear layer on top. The data for training the value function is collected from the base policy with a sampling temperature of 0.7, and we label it with the corresponding reward model.

实验设置。对于摘要和多轮对话任务，我们首先通过监督微调预训练语言模型，在相应的数据集上建立一个基础策略，以获得基本的任务特定能力和期望的行为。对于指令跟随任务，我们直接使用公开可用的指令微调模型。在所有实验中，我们将价值函数参数化为一个带有顶部线性层的预训练语言模型主干。用于训练价值函数的数据是从基础策略中以采样温度 0.7 收集的，并使用相应的奖励模型进行标注。

For IVR, we employ Value Exploration by sampling 4 different trajectories for each prompt to obtain robust value estimates. The training process involves two iterations of value function optimization to achieve better policy alignment. Starting from the second iteration, we collect training data using value-guided sampling with $\beta = 2$. More details can be found in Appendix B.

对于 IVR，我们通过为每个提示采样 4 条不同的轨迹来执行价值探索，以获得稳健的价值估计。训练过程包括两次价值函数优化迭代，以实现更好的策略对齐。从第二次迭代开始，我们使用 $\beta = 2$ 进行价值引导的采样来收集训练数据。更多细节请参见附录 B。

Evaluation Metrics. We adopt different evaluation metrics for our two decoding-time alignment strategies. For value-guided sampling, following (Gao et al., 2023; Han et al., 2024), we analyze the trade-off between reward and token-level KL divergence from the base policy. Specifically, we sweep the β in Equation 5 to control the KL divergence between the guided policy and base policy. For value guided blockwise beam search, we compare the reward of each algorithm. Additionally, as a supplementary evaluation to mitigate potential reward hacking issues (Amodei et al., 2016), we compute the win-rate between the guided policy and base policy using GPT-4-as-the-judge (Zheng et al., 2023a). We note that this serves as a complementary sanity check, while our primary conclusions are drawn from the objective reward and KL divergence metrics. The prompting template refers to Appendix C. To ensure the robustness of our evaluation, all experiments are conducted with 5 different random seeds.

评估指标。我们为两种解码时对齐策略采用不同的评估指标。对于值引导采样，我们遵循（Gao 等，2023；

Han 等, 2024) 的方法, 从基础策略的角度分析奖励与 token 级 KL 散度之间的权衡。具体而言, 我们遍历方程 5 中的 β 来控制引导策略与基础策略之间的 KL 散度。对于值引导块状束搜索, 我们比较每种算法的奖励。此外, 为了缓解潜在的奖励黑客问题 (Amodei 等, 2016), 我们使用 GPT-4 作为裁判计算引导策略与基础策略之间的胜率 (Zheng 等, 2023a)。我们注意到, 这仅作为补充的合理性检查, 而我们的主要结论来自于目标奖励和 KL 散度指标。提示模板参见附录 C。为确保评估的鲁棒性, 所有实验均使用 5 个不同的随机种子进行。

4.1 Experiment 1: Summarization

4.1 实验 1: 摘要生成

Experiment Details. We conduct experiments on the TL;DR dataset (Stiennon et al., 2020), which consists of Reddit posts paired with two candidate summaries and human preferences between them. For efficiency, we randomly sampled 300 examples from the test set for evaluation. For the base policy, we fine-tune a Llama-3.2-3B (Dubey et al., 2024) model on the preferred summaries using supervised learning. To evaluate summary quality, we train a reward model using a Llama-3.2-1B (Dubey et al., 2024) backbone on the pairwise preference data through Bradley-Terry (BT) modeling. The value function is implemented as a Llama-3.2-1B model with an additional linear layer on top. More implementation details can be found in Appendix B.

实验细节. 我们在 TL;DR 数据集 (Stiennon 等, 2020) 上进行实验, 该数据集包含 Reddit 帖子及其两个候选摘要和人类偏好。为了提高效率, 我们在测试集上随机选取了 300 个示例进行评估。对于基础策略, 我们使用监督学习在优选摘要上微调 Llama-3.2-3B (Dubey 等, 2024) 模型。为了评估摘要质量, 我们使用 Llama-3.2-1B (Dubey 等, 2024) 作为主干, 在成对偏好数据上通过 Bradley-Terry (BT) 建模训练奖励模型。价值函数实现为一个带有额外线性层的 Llama-3.2-1B 模型。更多实现细节请参见附录 B。

Baselines. We compare our method against several recent decoding-time alignment approaches. ARGS (Khanov et al., 2024) directly leverages reward models as value functions for guided decoding without additional training, offering a lightweight solution. Using trajectories sampled from the base policy π_{base} , FUDGE (Yang and Klein, 2021; Mudgal et al., 2023) trains a prefix scorer to predict future attributes, while VAS (Han et al., 2024) employs TD(λ) learning to train

基线。我们将我们的方法与几种最近的解码时对齐方法进行比较。ARGS (Khanov 等, 2024) 直接利用奖励模型作为引导解码的价值函数, 无需额外训练, 提供了一种轻量级的解决方案。使用从基础策略 π_{base} 中采样的轨迹, FUDGE (Yang 和 Klein, 2021; Mudgal 等, 2023) 训练一个前缀评分器以预测未来属性, 而 VAS (Han 等, 2024) 则采用 TD(λ) 学习进行训练

a value function, providing a more sophisticated value estimation approach.

一个价值函数, 提供一种更为精细的价值估计方法。

Results. As shown in Figure 2a, we analyze the trade-off between reward and KL divergence for different methods on the summarization task. IVR consistently outperforms all baselines across different KL divergence levels, achieving higher rewards while maintaining stable performance. Specifically, at KL divergence of 0.8, IVR reaches a reward of approximately 3.5, while other methods remain below 3.2. In contrast, baseline methods (FUDGE, VAS, and ARGS) show performance degradation when KL divergence exceeds 0.4, suggesting their limited capability in balancing policy preservation and performance optimization.

结果. 如图 2a 所示, 我们分析了在摘要任务中, 不同方法在奖励与 KL 散度之间的权衡。IVR 在不同 KL 散度水平下始终优于所有基线方法, 在保持性能稳定的同时获得更高的奖励。具体而言, 在 KL 散度为 0.8 时, IVR 的奖励约为 3.5, 而其他方法则保持在 3.2 以下。相比之下, 基线方法 (FUDGE、VAS 和 ARGS) 在 KL 散度超过 0.4 时表现出性能下降, 这表明它们在平衡策略保持和性能优化方面的能力有限。

As shown in Figure 3a, all methods demonstrate improvements over the base policy (3.2) in value-guided blockwise beam search, with our IVR method achieving the highest performance (4.3). Specifically, ARGS yields a modest improvement to 3.55, while FUDGE and VAS demonstrate

stronger performance at 4.05 and 4.15 respectively. The ablation of our method without Iterative Self-Refinement (IVR w/o Iter) achieves 4.25, highlighting the effectiveness of our Value Exploration approach. The full IVR method further improves the performance to 4.3, demonstrating the benefits of Iterative Self-Refinement.

如图 3a 所示，在基于价值引导的块状束搜索中，所有方法均在基线策略（3.2）的基础上表现出提升，其中我们的 IVR 方法取得了最高的性能（4.3）。具体而言，ARGS 方法实现了适度的提升，达到 3.55，而 FUDGE 和 VAS 分别表现出更强的性能，分别为 4.05 和 4.15。我们的方法在不使用迭代自优化（IVR w/o Iter）时的消融实验结果为 4.25，突显了我们的价值探索方法的有效性。完整的 IVR 方法进一步将性能提升至 4.3，展示了迭代自优化的优势。

4.2 Experiment 2: Multi-turn Dialogue

4.2 实验 2：多轮对话

Experiment Details. We use the Anthropic HH (Bai et al., 2022) dataset, a multi-turn dialogue dataset focused on helpfulness and harmlessness, sampling 300 examples for evaluation. We train the base policy by fine-tuning Llama-3-8B (Dubey et al., 2024) on preferred responses. A Llama-3.2-1B model is trained as the reward model using BT on pairwise preference data. Llama-3.2-1B serves as the value function backbone. More details are in Appendix B.

实验细节。 我们使用 Anthropic HH (Bai 等, 2022) 数据集，这是一个专注于有用性和无害性的多轮对话数据集，采样 300 个示例用于评估。我们通过在偏好响应上微调 Llama-3-8B (Dubey 等, 2024) 来训练基础策略。使用 BT 在成对偏好数据上训练了一个 Llama-3.2-1B 模型作为奖励模型。Llama-3.2-1B 作为价值函数的主干。更多细节请参见附录 B。

Baselines. In addition to the aforementioned inference-based baselines (ARGS, FUDGE, and VAS), we also include several training-based baselines: Direct Preference Optimization (DPO) (Rafailov et al., 2024) and Identity Preference Optimization (IPO) (Azar et al., 2024). DPO directly fine-tunes the model for preference learning, eliminating the need for a reward model and RL stage for updates. IPO added a regularization term to the DPO objective to mitigate overfitted risk. We used the online version of DPO and IPO by rolling

基线。 除了上述基于推理的基线（ARGS、FUDGE 和 VAS）之外，我们还包含了一些基于训练的基线：直接偏好优化（DPO）(Rafailov 等人, 2024) 和身份偏好优化（IPO）(Azar 等人, 2024)。DPO 直接对模型进行偏好学习的微调，消除了对奖励模型和 RL 阶段进行更新的需要。IPO 在 DPO 目标中添加了一个正则化项以减轻过拟合的风险。我们通过滚动使用 DPO 和 IPO 的在线版本来实现这一点。

Methods	Win-Rate (%)	FLOPs ($\times 10^{15}$)
FUDGE	64.85	0.25
VAS	68.49	0.25
DPO	72.45	4.00
IPO	66.55	4.00
IVR (Ours)	77.52	2.00

方法	胜率 (%)	FLOPs ($\times 10^{15}$)
FUDGE	64.85	0.25
VAS	68.49	0.25
DPO	72.45	4.00
IPO	66.55	4.00

IVR (我们的)	77.52	2.00
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Table 1: Comparison of different methods’ win-rate against the base policy and estimated FLOPs. The win-rate values are obtained by GPT-4-as-the-judge on the Multi-turn Dialogue dataset. FLOPs are estimated on a single batch training.

表 1: 不同方法与基线策略的胜率对比以及估计的 FLOPs。胜率值由 GPT-4 作为裁判在轮对话数据集上获得。FLOPs 是基于单个批次训练进行估计的。

out the base policy and sampling two trajectories, optimizing objective on explicit rewards.

输出基础策略并抽样两条轨迹，优化目标在显式奖励上。

Results. As demonstrated in Figure 2b, we evaluate different methods on the multi-turn dialogue task. Our method achieves the best performance across different KL divergence levels, reaching a reward of 1.75 at KL divergence of 0.3. DPO shows competitive performance initially but plateaus at a reward of 1.65, while VAS and FUDGE demonstrate moderate performance with rewards of 1.5 and 1.3 respectively. ARGS and IPO show limited effectiveness, with ARGS achieving minimal improvement and IPO’s performance degrading significantly as KL divergence increases. Notably, IVR maintains stable performance even at higher KL divergence levels (0.4-0.6), while other methods either plateau or decline.

结果。如图 2b 所示，我们在多轮对话任务上评估了不同的方法。我们的方法在不同 KL 散度水平下均表现最佳，在 KL 散度为 0.3 时达到 1.75 的奖励值。DPO 在初期表现出竞争力，但在奖励值为 1.65 时趋于平稳，而 VAS 和 FUDGE 则表现出中等水平的性能，分别获得 1.5 和 1.3 的奖励值。ARGS 和 IPO 表现出有限的效果，其中 ARGS 仅获得微小的改进，而 IPO 的性能随着 KL 散度的增加显著下降。值得注意的是，IVR 即使在较高的 KL 散度水平（0.4-0.6）下也能保持稳定的性能，而其他方法则要么趋于平稳，要么下降。

For value-guided blockwise beam search in Figure 3b, we observe a clear progression in performance across different methods. Starting from the base policy, ARGS provides initial improvements through direct reward model utilization, achieving a reward of 2.1. VAS and FUDGE demonstrate stronger performance at around 2.7, while the ablation of our method without iterative training (IVR w/o Iter) reaches 2.8. The full IVR method with iterative training achieves the best performance with a reward of 3.1. The significant performance gap between IVR and IVR w/o Iter (0.3) further validates the effectiveness of our iterative training strategy.

在图 3b 中，对于基于价值引导的块状束搜索，我们观察到不同方法之间性能的明显提升。从基础策略开始，ARGS 通过直接利用奖励模型提供了初始改进，达到了 2.1 的奖励值。VAS 和 FUDGE 在大约 2.7 的水平上表现出更强的性能，而我们的方法在没有迭代训练（IVR w/o Iter）的消融实验中达到了 2.8。带有迭代训练的完整 IVR 方法取得了最佳性能，奖励值为 3.1。IVR 与 IVR w/o Iter 之间显著的性能差距（0.3）进一步验证了我们迭代训练策略的有效性。

Additionally, as a supplementary evaluation, Table 1 shows that our IVR method achieves the highest GPT-4 win-rate of 77.52%, corroborating the reward-based findings above. IVR also demonstrates a significant advantage in computational efficiency, with FLOPs estimated at 2.00×10^{15} , which is half of that required by DPO and IPO under our experimental settings.

此外，作为补充评估，表 1 显示我们的 IVR 方法实现了最高的 GPT-4 胜率 77.52%，这证实了上述基于奖励的发现。IVR 在计算效率方面也表现出显著优势，其计算量估计为 2.00×10^{15} ，这在我们的实验设置下仅为 DPO 和 IPO 所需的一半。

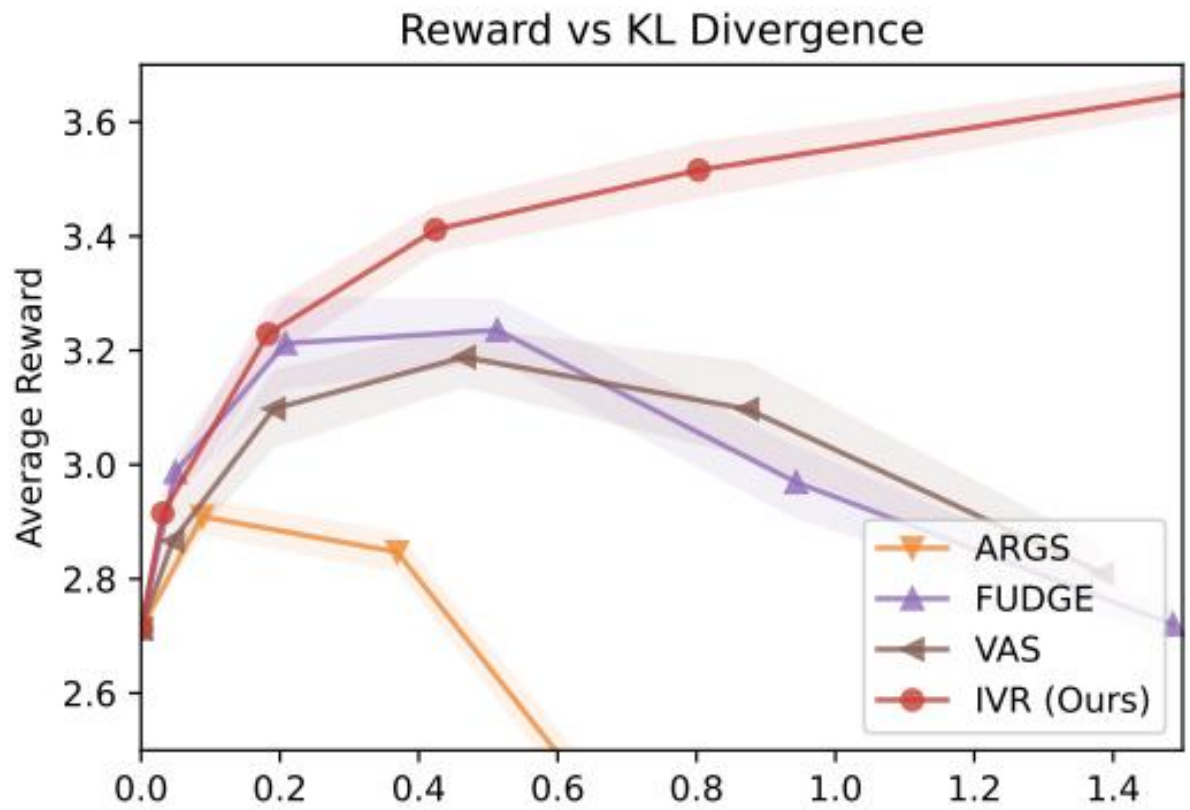


Figure 3: images/3f392bc7b6bbfe997f0da42946b2b40710f7b376f7b69fd74971fc19e6771630.jpg

(a) Summarization

(a) 概括

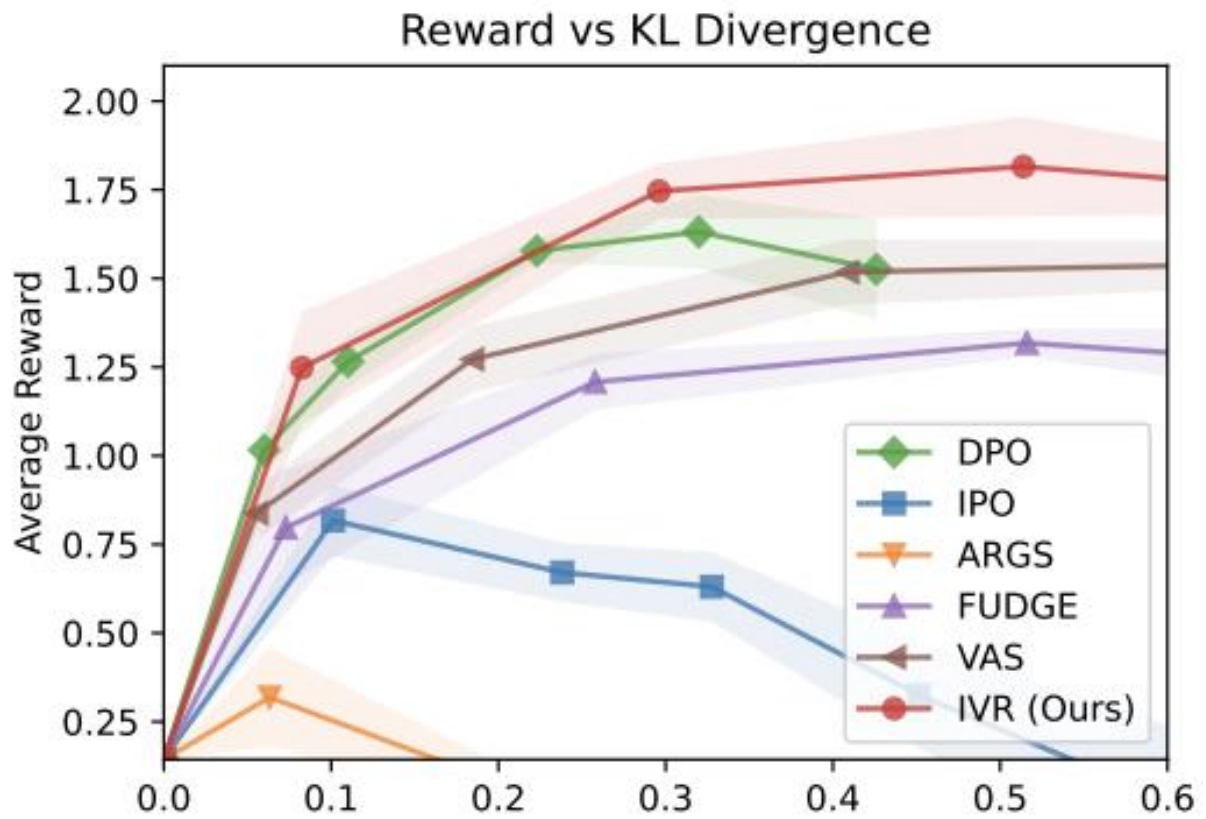


Figure 4: images/a8b2024b1d65e5f0cce1c9da809f9e1f04f87fc6d6109876ac54d389d5984b87.jpg

(b) Multi-turn Dialogue

(b) 多轮对话

Figure 2: Reward vs. KL divergence for different methods on (a) summarization and (b) multi-turn dialogue.

图 2：不同方法在 (a) 摘要任务和 (b) 多轮对话任务上的奖励与 KL 散度。

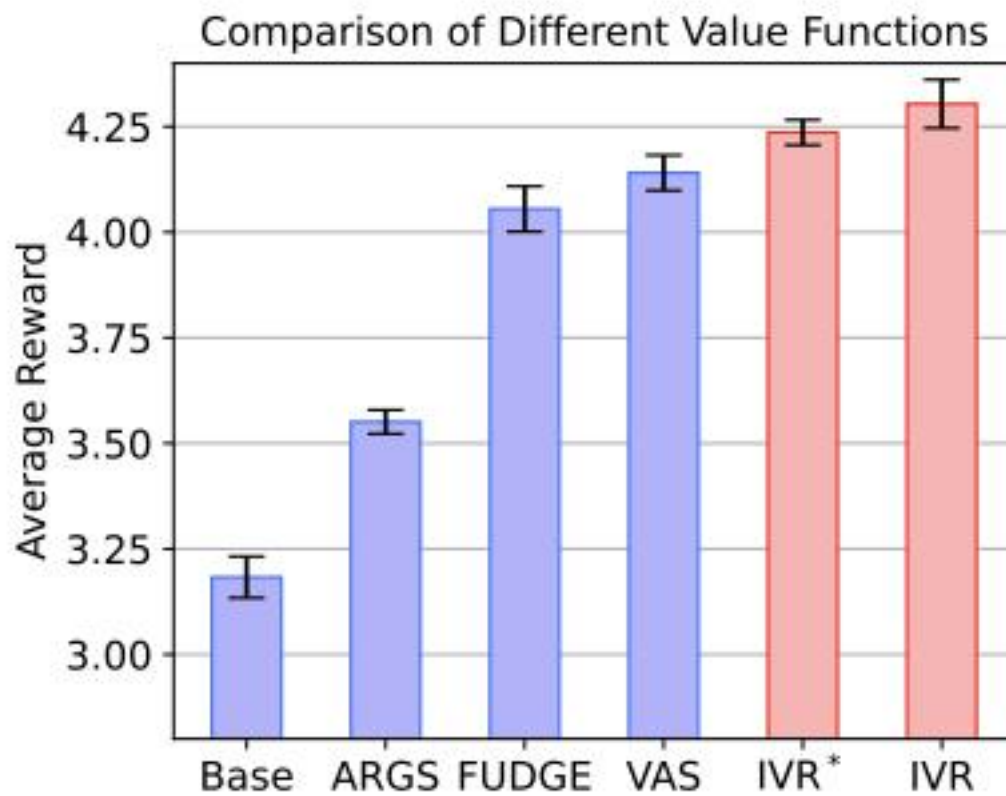


Figure 5: images/4a06f0ef77e37dd5302c8a33e7f631bfb513f245c38b0dfc9d6736a94e14189c.jpg

(a) Summarization

(a) 概括总结

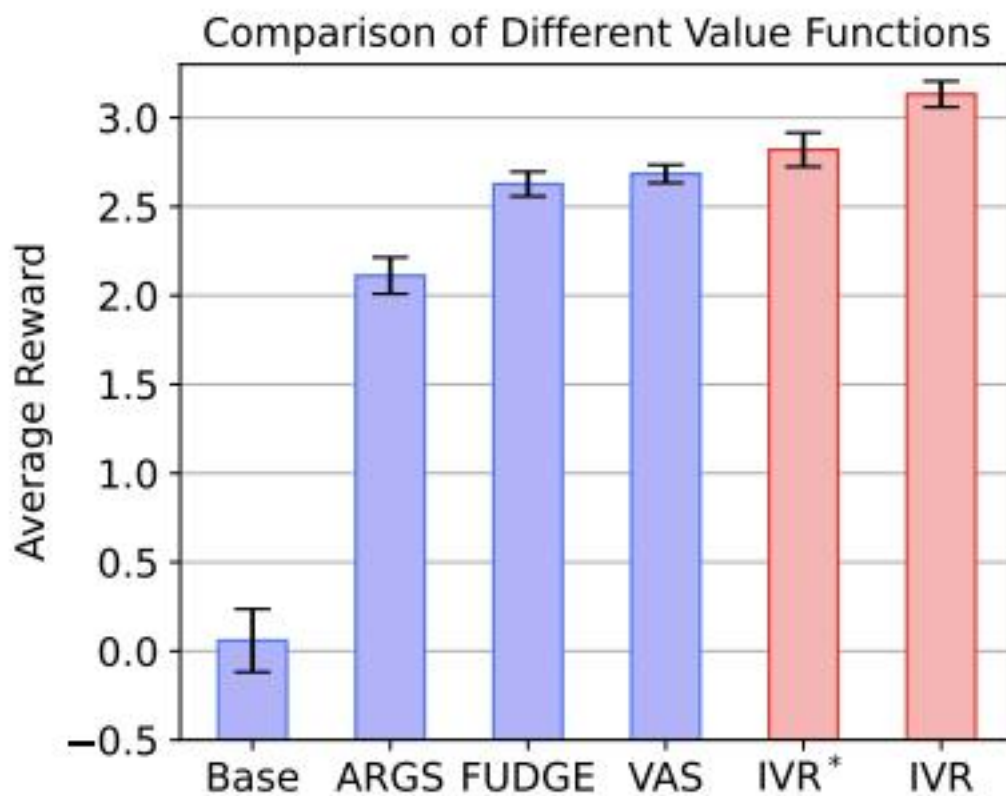


Figure 6: images/f700781b61791388d862abebbb497590e08327c27791a5b062e0b500620f747d8.jpg

(b) Multi-turn Dialogue

(b) 多轮对话

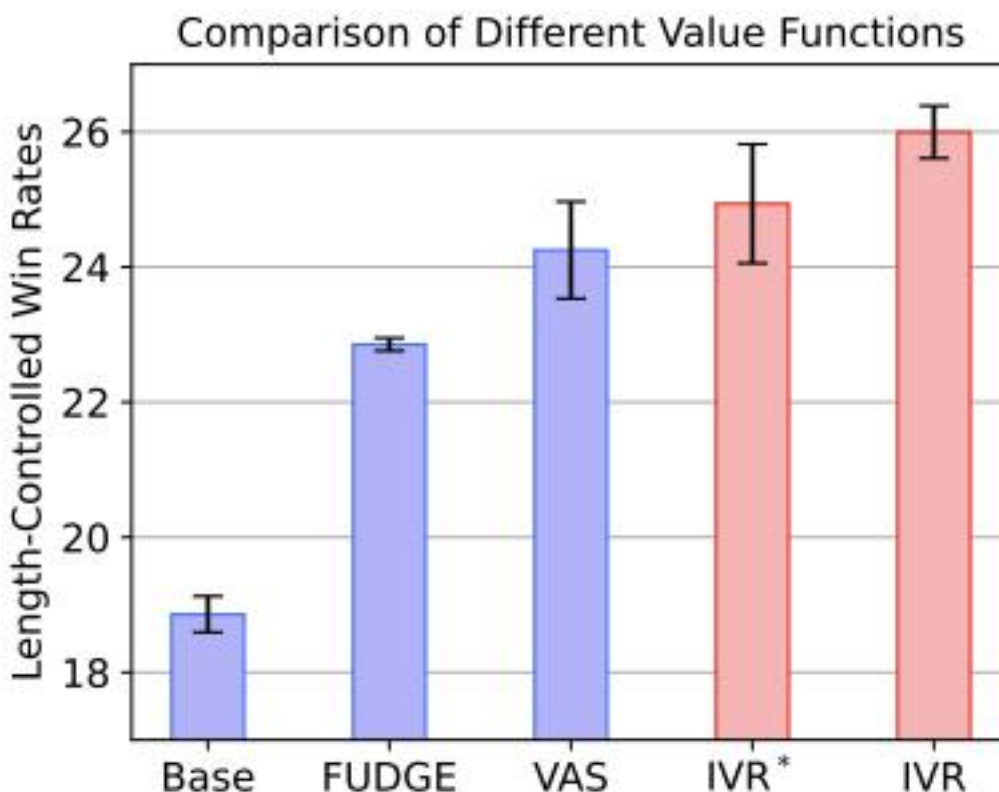


Figure 7: images/633e093e99a9999c91c20eb8e653785d77380ffa76ee4ae9b3bb5103894a68f8.jpg

(c) Instruction Following

Figure 3: Comparison of different value functions using value-guided blockwise beam search on summarization (left), multi-turn dialogue (middle) and instruction following (right). IVR* denotes IVR without Iterative Self-Refinement.

(c) 指令遵循

图 3: 在摘要（左）、多轮对话（中）和指令遵循（右）任务中，使用基于价值引导的分块束搜索比较不同价值函数的性能。IVR* 表示不包含迭代自我优化的 IVR。

4.3 Experiment 3: Instruction Following

4.3 实验 3：遵循指示

Experiment Details. We conduct experiments using the UltraFeedback (Cui et al., 2023) dataset with 10k sampled prompts for training. For evaluation, we use Alpaca-Eval 2 (Dubois et al., 2024), which assesses instruction-following via Length-controlled Win Rate with GPT-4 (Achiam et al., 2023). The base policy is Llama-3.2-3B-Instruct (Dubey et al., 2024), with Skywork-Reward-Llama-3.1-8B (Liu et al., 2024b) for reward modeling and Llama-3.2-1B-Instruct (Dubey et al., 2024) as the value function. For efficiency, we compare against FUDGE and VAS as decoding-time baselines. More details are in Appendix B.

实验细节。我们使用 UltraFeedback (Cui 等, 2023) 数据集进行实验, 该数据集包含 10,000 个采样的提示用于训练。在评估方面, 我们使用 Alpaca-Eval 2 (Dubois 等, 2024), 它通过 GPT-4 (Achiam 等, 2023) 的长度控制胜率来评估指令跟随能力。基础策略是 Llama-3.2-3B-Instruct (Dubey 等, 2024), 使用 Skywork-Reward-Llama-3.1-8B (Liu 等, 2024b) 进行奖励建模, 并使用 Llama-3.2-1B-Instruct (Dubey 等, 2024) 作为价值函数。为了提高效率, 我们与 FUDGE 和 VAS 作为解码时间基线进行比较。更多细节请参见附录 B。

Results. For the instruction following task, we evaluate our method using Length-controlled Win Rate as the metric. As shown in Figure 3c, all methods demonstrate improvements over the base policy. Our IVR method achieves the best performance with a win rate of 26.0%, significantly outperforming both FUDGE and VAS. The consistent improvements across different methods indicate the effectiveness of value-guided decoding for instruction following. Notably, the performance gap between IVR and other methods suggests that our iterative value optimization approach is particularly beneficial for complex tasks like instruction following, where accurate value estimation is crucial for generating high-quality responses.

结果。对于指令跟随任务，我们使用长度控制胜率（Length-controlled Win Rate）作为评估指标。如图 3c 所示，所有方法在基线策略的基础上都取得了改进。我们的 IVR 方法表现最佳，胜率为 26.0%，显著优于 FUDGE 和 VAS。不同方法之间的一致性改进表明，基于价值引导的解码对于指令跟随任务是有效的。值得注意的是，IVR 与其他方法之间的性能差距表明，我们的迭代价值优化方法对于像指令跟随这样的复杂任务特别有益，在这种任务中，准确的价值估计对于生成高质量的响应至关重要。

5 Further Analysis

5 进一步分析

This section analyzes IVR's key components, examining the impact of Value Exploration and training iterations on performance. We also evaluate the transferability of value functions across model scales and IVR's effectiveness in enhancing model safety against adversarial jailbreak attacks, demonstrating its broader applications.

本节分析 IVR 的关键组成部分，探讨价值探索和训练迭代对性能的影响。我们还评估价值函数在不同模型规模之间的迁移性，以及 IVR 在增强模型对抗对抗性越狱攻击方面的有效性，展示了其更广泛的应用。

5.1 The Number of Sampled Trajectories

5.1 采样轨迹的数量

To investigate the impact of Value Exploration, we analyze how the number of sampled trajectories affects value function performance. We conduct experiments on the multi-turn dialogue task using value-guided blockwise beam search, without Iterative Self-Refinement. The results are shown in Figure 4a. We observe that increasing the num-

为了研究价值探索的影响，我们分析采样轨迹数量如何影响价值函数的性能。我们在多轮对话任务上进行实验，使用价值引导的块状束搜索方法，不使用迭代自优化。结果如图 4a 所示。我们观察到，增加采样轨迹数量会...

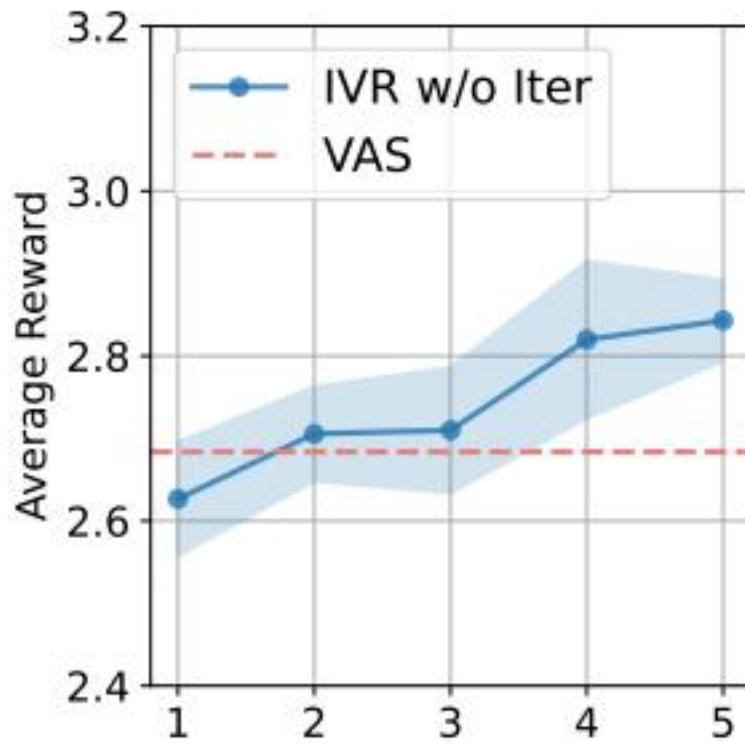


Figure 8: images/b1ddc60629c9f12f3df4a92896894ca3d320e853b3a29c405e3dc44cf9b8bae5.jpg

(a) Number of ST

(a) ST 数量

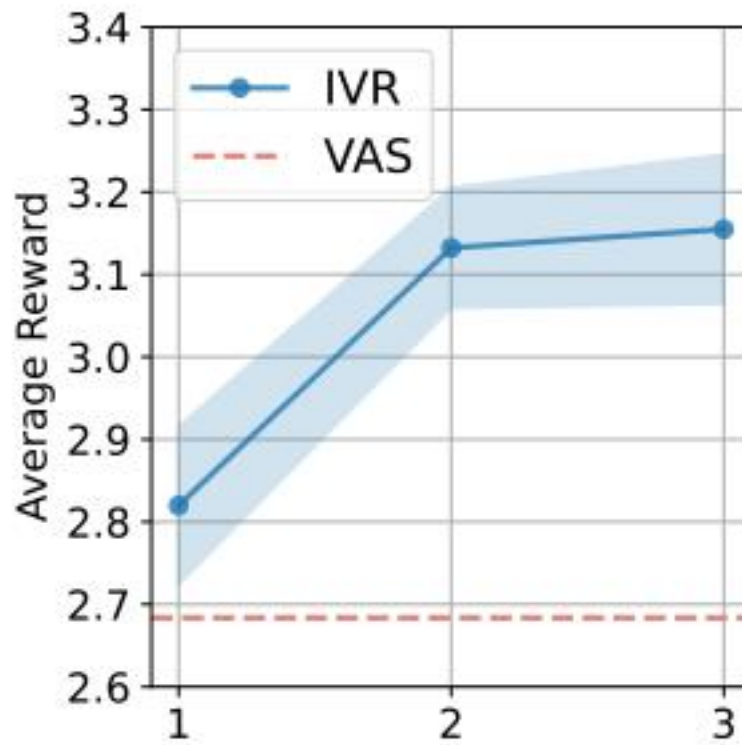


Figure 9: images/4fbfb16107407231e518b1a9cacd331f90ec604216ef34b94058f82dfe4cf44d.jpg

(b) Number of TI

Figure 4: Ablation studies on (a) the number of sampled trajectories and (b) the number of training iterations in multi-turn dialogue using blockwise beam search. ST denotes sampled trajectories, TI denotes training iterations.

(b) 训练迭代次数

图 4：使用分块束搜索在多轮对话中对（a）采样轨迹数量和（b）训练迭代次数进行消融研究。ST 表示采样轨迹，TI 表示训练迭代次数。

ber of sampled trajectories leads to consistent improvements. A more significant improvement is observed with 4 samples. Further increasing to 5 samples only brings a small additional improvement, suggesting that 4 trajectories provide a good balance between computational cost and performance. These results demonstrate that Value Exploration with multiple trajectories helps capture a more comprehensive view of possible outcomes, leading to more accurate value estimates and better guided generation.

采样的轨迹数量导致了一致的改进。使用 4 个样本时观察到更显著的改进。进一步增加到 5 个样本仅带来小幅的额外改进，这表明 4 条轨迹在计算成本和性能之间提供了良好的平衡。这些结果表明，使用多条轨迹的值探索有助于捕捉更全面的可能结果视图，从而获得更准确的价值估计和更好的引导生成。

5.2 The Number of Training Iterations

5.2 训练迭代次数

We analyze how the number of training iterations affects model performance. We conduct experiments on the multi-turn dialogue task using value-guided blockwise beam search, with the number of sampled trajectories fixed at 4 based on our previous findings. The results are shown in Figure 4b. Starting from one iteration, our method already outperforms VAS with an average reward of 2.82. The second iteration brings a substantial improvement, demonstrating the effectiveness of collecting training data from the guided policy. The third iteration yields a slight gain to 3.15, suggesting that two iterations provide sufficient policy alignment.

我们分析了训练迭代次数对模型性能的影响。我们基于之前的研究结果，将采样轨迹数固定为 4，在多轮对话任务中使用价值引导的块状束搜索方法进行实验。实验结果如图 4b 所示。从一次迭代开始，我们的方法就已经在平均奖励上超过了 VAS，达到了 2.82。第二次迭代带来了显著的性能提升，这表明从引导策略中收集训练数据是有效的。第三次迭代仅带来小幅提升，达到 3.15，这表明两次迭代已经提供了足够的策略对齐。

5.3 Value Function Transferability Across Model Sizes

5.3 不同模型规模之间的价值函数可迁移性

We investigate the transferability of estimated value function across different model sizes on the instruction following task. Specifically, we examine whether a value function trained using data collected from a Llama-3.2-3B-Instruct can effectively guide models of different sizes (Llama-3.2-1B-Instruct and Llama-3-8B-Instruct) during value-guided blockwise beam search.

我们研究了估计的价值函数在不同模型规模上的迁移性，特别是在遵循指令任务中的表现。具体而言，我们考察了使用从 Llama-3.2-3B-Instruct 收集的数据训练的价值函数，是否能够有效指导不同规模的模型（如 Llama-3.2-1B-Instruct 和 Llama-3-8B-Instruct）在价值引导的分块束搜索中进行。

Model Size	Base	IVR	Δ
1B	-7.71	3.78	11.49
3B	1.56	13.34	11.78

8B	7.67	20.42	12.75
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模型规模	基础	IVR	Δ
1B	-7.71	3.78	11.49
3B	1.56	13.34	11.78
8B	7.67	20.42	12.75

Table 2: Performance comparison between the base policy and value-guided blockwise beam search across different model sizes (1B, 3B, and 8B) on the instruction-following task. The value function is trained using data collected from the 3B base policy. The Δ column represents the absolute performance improvement.

表 2：在指令遵循任务中，基线策略与值引导的块状束搜索在不同模型规模（1B、3B 和 8B）上的性能比较。值函数使用从 3B 基线策略收集的数据进行训练。 Δ 列表示绝对性能提升。

Model	Safety Rate (%)
Aligned-Llama-3	92.87
+ Aligned Value Function	96.14
+ Unaligned Value Function	97.48
Unaligned-Llama-3	65.27
+ Aligned Value Function	81.04
+ Unaligned Value Function	86.17

模型	安全率 (%)
对齐的 Llama-3	92.87
+ 对齐的价值函数	96.14
+ 非对齐价值函数	97.48
Unaligned-Llama-3	65.27
+ 对齐的价值函数	81.04
+ 非对齐价值函数	86.17

Table 3: Comparison of safety rates between aligned and unaligned Llama-3 models with different value functions.

表 3：对齐与未对齐的 Llama-3 模型在不同价值函数下的安全率对比。

As shown in Table 2, our value function demonstrates strong transferability across model scales. These results suggest that the value function learned through IVR captures generalizable knowledge about task-specific preferences that can be applied to guide models of varying sizes. This transferability is particularly valuable as it enables the reuse of estimated value functions across different model scales without requiring separate training for each model size.

如表 2 所示，我们的价值函数在模型规模上表现出强大的可迁移性。这些结果表明，通过 IVR 学习到的价值函数捕捉了关于任务特定偏好的可推广知识，可以用于指导不同规模的模型。这种可迁移性尤其有价值，因为它使得可以在不同模型规模之间重用估计的价值函数，而无需为每个模型规模单独进行训练。

5.4 Enhancing Safety Against Jailbreak Attacks

5.4 提升对越狱攻击的安全性

To investigate whether IVR can enhance instruction-tuned model safety against jailbreak attacks, we conduct experiments using Llama-3-8B-Instruct as an aligned version and Llama-3-8B-Lexi-Uncensored¹ as an unaligned version. We evaluate their safety rate against jailbreak attacks using the attack-enhanced set split from the SALAD-Bench (Li et al., 2024), which contains 5,000 harmful questions generated through multiple jailbreak attack methods. We use 4,800 prompts for training and 200 for testing. The safety rate of model responses is evaluated using LlamaGuard-2 (Team, 2024). To comprehensively evaluation, we first obtain aligned and unaligned

为了调查 IVR 是否能够增强指令调优模型在对抗越狱攻击方面的安全性，我们使用 Llama-3-8B-Instruct 作为对齐版本，Llama-3-8B-Lexi-Uncensored¹ 作为非对齐版本进行实验。我们采用 SALAD-Bench (Li 等, 2024) 中通过攻击增强的数据集分割方式来评估其对抗越狱攻击的安全性，该数据集包含 5,000 个通过多种越狱攻击方法生成的有害问题。我们使用 4,800 个提示进行训练，200 个进行测试。模型响应的安全性评估使用 LlamaGuard-2 (Team, 2024) 进行。为了全面评估，我们首先获取对齐和非对齐

value functions by applying IVR on the aligned and unaligned models respectively. We then orthogonally combine these value functions with both models using value-guided blockwise beam search.

通过分别在对齐和未对齐的模型上应用 IVR 来获得价值函数。然后，我们使用以价值为导向的块状束搜索方法，将这些价值函数与两个模型正交地结合。

As shown in Table 3, IVR significantly improves safety rates for both aligned and unaligned models through value-guided blockwise beam search. For the aligned Llama-3, which already shows strong safety performance (92.87%), IVR improves its safety rate to 96.14% and 97.48% with aligned and unaligned value functions, respectively. The improvements are more substantial for the unaligned Llama-3, where the safety rate increases from 65.27% to 81.04% with aligned value function and 86.17% with unaligned value function. Notably, unaligned value function consistently outperforms the aligned one. We hypothesize that the reason is the unaligned value function is trained on responses from the unaligned model, which explores a more diverse solution space due to its lack of alignment. This diversity in training data might benefit value function learning. We leave further investigation of this hypothesis to future work.

如表 3 所示，通过价值引导的块状束搜索，IVR 显著提高了对齐和未对齐模型的安全性指标。对于已经表现出较强安全性表现 (92.87%) 的对齐 Llama-3 模型，IVR 分别通过对齐和未对齐的价值函数将其安全性指标提升至 96.14% 和 97.48%。对于未对齐的 Llama-3 模型，其安全性指标的提升更为显著，使用对齐价值函数时从 65.27% 提升至 81.04%，使用未对齐价值函数时则提升至 86.17%。值得注意的是，未对齐的价值函数表现始终优于对齐的价值函数。我们假设其原因在于，未对齐的价值函数是在未对齐模型的响应上进行训练的，由于其缺乏对齐，因此探索了更加多样化的解决方案空间。这种训练数据的多样性可能有助于价值函数的学习。我们留待未来的工作进一步验证这一假设。

6 Related Work

6 相关工作

Reinforcement Learning for Language Models. Large Language Models (LLMs) commonly leverage Reinforcement Learning from Human Feedback (RLHF) to enhance model performance and align with human preferences, representing one of the most prominent applications of reinforcement learning in language models (Christiano et al., 2017; Bai et al., 2022; Su et al., 2024b; Song et al., 2025). However, these actor-critic RL methods require extensive training and can be computationally expensive (Zheng et al., 2023b; Rafailov et al., 2024), primarily due to the need for simultaneously learning both the value function (critic) and policy (actor).

语言模型中的强化学习。大型语言模型（LLMs）通常利用来自人类反馈的强化学习（RLHF）来提升模型性能并使其符合人类偏好，这代表了强化学习在语言模型中最突出的应用之一（Christiano 等，2017；Bai 等，2022；Su 等，2024b；Song 等，2025）。然而，这些 actor-critic 强化学习方法需要大量的训练，并且计算成本较高（Zheng 等，2023b；Rafailov 等，2024），主要原因是需要同时学习价值函数（critic）和策略（actor）。

Guided Decoding. Guided decoding represents a family of techniques that steer language model outputs at inference time while keeping model parameters frozen, offering both efficiency and flexibility compared to traditional RLHF methods. Huang et al. (2024) views the decoding process as a heuristic-guided search problem. Some works utilize contrastive decoding methods by combining distributions from multiple sources, typically using either prompting strategies (Dekoning et al.; Zhong et al., 2024) or training small models (Liu et al., 2021, 2024a). Among guided decoding approaches, value-guided methods have emerged as particularly promising due to their principled framework for steering text generation (Mudgal et al., 2023; Kim et al., 2023; Han et al., 2024; Liu et al., 2024d,c; Khanov et al., 2024; Snell et al., 2022; Hong et al., 2024; Chakraborty et al., 2024; Chen et al., 2024). In contrast, our approach explicitly targets the distributional gap that limits prior value-guided methods. While ARGS (Khanov et al., 2024) bypasses value training and relies directly on a reward model, FUDGE (Yang and Klein, 2021; Mudgal et al., 2023) and VAS (Han et al., 2024) train value functions on a fixed pool of base-policy trajectories, leaving them blind to regions of output space that the optimal policy actually visits. IVR closes this gap through Value Exploration and Iterative Self-Refinement, while avoiding the full policy retraining required by preference-based methods such as DPO (Rafailov et al., 2024) and IPO (Azar et al., 2024).

引导解码。引导解码代表了一类在推理过程中引导语言模型输出的技术，同时保持模型参数不变，相较于传统的 RLHF 方法，提供了更高的效率和灵活性。Huang 等人（2024）将解码过程视为一种启发式引导的搜索问题。一些研究通过结合多个来源的分布来利用对比解码方法，通常采用提示策略（Dekoning 等人；Zhong 等人，2024）或训练小型模型（Liu 等人，2021, 2024a）。在引导解码方法中，基于价值引导的方法因其在文本生成方面的原则性框架而显得尤为有前景（Mudgal 等人，2023；Kim 等人，2023；Han 等人，2024；Liu 等人，2024d,c；Khanov 等人，2024；Snell 等人，2022；Hong 等人，2024；Chakraborty 等人，2024；Chen 等人，2024）。相比之下，我们的方法明确针对限制了先前基于价值引导方法的分布差距。虽然 ARGS（Khanov 等人，2024）绕过了价值训练，直接依赖于奖励模型，但 FUDGE（Yang 和 Klein, 2021；Mudgal 等人，2023）和 VAS（Han 等人，2024）则在固定的基础策略轨迹池上训练价值函数，使它们无法察觉最优策略实际访问的输出空间区域。IVR 通过价值探索和迭代自优化来弥合这一差距，同时避免了基于偏好的方法（如 DPO（Rafailov 等人，2024）和 IPO（Azar 等人，2024））所需的完整策略再训练。

7 Conclusion

7 结论

This paper introduces IVR, a novel framework for guided decoding that addresses key limitations in value-guided approaches through Value Exploration and Iterative Self-Refinement. Our extensive experiments across text summarization, multi-turn dialogue, and instruction following demonstrate that IVR consistently outperforms existing methods. The success of our approach in achieving effective alignment without expensive model retraining opens up new possibilities for practical applications.

本文介绍了 IVR，一个新颖的引导解码框架，通过价值探索和迭代自优化解决了价值引导方法中的关键限制。我们在文本摘要、多轮对话和指令遵循等任务上的广泛实验表明，IVR 始终优于现有方法。我们的方法在无需昂贵模型重新训练的情况下实现有效对齐的成功，为实际应用开辟了新的可能性。

Limitations

限制

Computational Overhead. Although IVR avoids the full policy retraining of RLHF, it still incurs additional inference-time cost from value function evaluations and additional training cost from iterative data collection. We mitigate the inference overhead via blockwise sampling (Appendix A), but the total training cost grows roughly linearly with the number of iterations.

计算开销。尽管 IVR 避免了 RLHF 的完整策略重新训练，但它仍会因价值函数评估而产生额外的推理时间成本，并因迭代数据收集而产生额外的训练成本。我们通过分块采样（附录 A）来缓解推理开销，但总的训练成本大致随着迭代次数线性增长。

Base Policy Dependence. Both value-guided sampling and blockwise beam search operate within the support of the base policy’s top- k distribution. This preserves coherence inherited from the base model, but precludes IVR from discovering high-reward outputs that lie entirely outside the base policy’s typical generations.

基础策略依赖性。值引导采样和分块束搜索操作均在基础策略的前 k 分布的支持范围内。这保留了来自基础模型的连贯性，但阻止了 IVR 发现完全超出基础策略典型生成的高奖励输出。

Acknowledgments

致谢

This work is supported by Shanghai Artificial Intelligence Laboratory.

本工作得到上海人工智能实验室的支持。

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